MIL-H-10063(ORD)
9 December 1949
SUPERSEDING
Specification WWES-148
18 May 1949

MILITARY SPECIFICATION

HOWITZER, 105-MM., T96

1. SCOPE

1.1 This specification covers one type of cannon, known as Howitzer, 105mm., T96.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.- The following specifications and specifications referenced on drawings in 2.2, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

TT-E-485 - Enamel; (For) Drums and Other Metal Products, Rust-Inhibiting, Olive Drab.

MILITARY SPECIFICATIONS

JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-P-116 - Packaging and Packing for Overseas Shipment - Preservation, Methods of.
JAN-B-121 - Barrier-Materials, Greaseproof.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-140 - Packaging and Packing for Overseas Shipment - Adhesives, Water-Resistant, Cast-Liner.

U. S. ARMY SPECIFICATION

57-206-1 - Welding of Constructional Steel, Metal-Arc, Manual or Machine; For Artillery and Small Arms Materiel: (Excluding Submarine Mines and Observation Towers).

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings.- The following drawing, and drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

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ARMY ORDNANCE DEPARTMENT DRAWING
52-298-1 Howitzer, 105mm, T96, List of Drawings and List of Specifications.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Material.- See detail drawings, material specifications, and paragraph titled "Material" in section III of Specification 51-1.

3.2 Workmanship.- See paragraphs titled "Workmanship" and "Methods of Manufacture" in section III of Specification 51-1.

3.3 Machining.- Extreme care is essential in machining the tube (rough turning, rough boring, resetting a sufficient number of times), to make sure that all stresses which may distort the cannon are removed prior to finish machining.

3.4 Bearing Surfaces.- Surfaces on which uniform contact or bearing is specified shall be spotted and hand scraped as necessary to obtain the required contact.

3.5 Welding.- Welding prescribed on the drawings shall comply with the applicable requirements of Specification 57-206-1.

3.6 Mechanical Requirements.

3.6.1 The pressure bearing surfaces of the breechblock shall show uniform bearing on the corresponding surfaces in the breech recess.

3.6.2 The extractors shall operate in unison, with substantially equal pressure on the flange of the cartridge case, to eject the empty case clear of the howitzer with the howitzer in horizontal position.

3.6.3 Overtravel of the firing pin guide in cocking, when measured as prescribed in paragraph 4.4., shall be not less than 0.07 inch.

3.6.4 The radii of all fillets shall comply with the limits specified on the applicable drawings, and the surface finish of all fillets shall comply with the finish requirement for adjacent surfaces.

3.7 Marking.- Howitzers shall be marked in accordance with applicable drawing, except that the manufacturers name, or an abbreviation thereof, shall appear in place of "WTV. ARS".

3.8 Painting.- Howitzers shall be painted in accordance with applicable requirements of the paragraph titled "Painting" in section IV of Specification 51-1, except that the priming coat and finish coat shall each consist of enamel, rust-inhibiting, olive drab, conforming to Specification TT-E-485.

3.9 Performance.- See paragraph titled "Assembly and Performance" in section IV of Specification 51-1.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 See section V of Specification 51-1.

4.2 Bearing Surfaces.- Surfaces for which uniform contact or bearing is specified shall be tested with proper gages, using prussian blue or equivalent for proof of contact.

4.3 Extraction and ejection shall be tested with an empty service cartridge case.

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4.4 Headspace and Firing Pin Protrusion.- Headspace, protrusion of the firing pin, and overtravel of the firing pin guide in cocking shall be measured. The overtravel of the firing pin guide shall be determined as follows:

Replace the standard firing spring retainer with a similar retainer having a hole from front to rear of such size as to admit a depth gage to the rear face of the firing pin guide. Lower the breechblock slowly to the point where the sear engages the firing pin guide, as indicated by an audible click. At this point, take a depth gage reading from the rear face of the breechblock to the rear face of the firing pin guide. Keeping the base of the depth gage in position against the breechblock, further lower the breechblock until the guide reaches its rearmost position, and take a second reading. The difference between the two readings is the measurement of overtravel.

4.5 Functioning Tests.

4.5.1 See paragraphs titled "Firing Mechanism" and "Maneuvering Test" in section V of Specification 51-1.

4.6 Proof Firing.

4.6.1 See paragraph titled "Proof Firing" in section V of Specification 51-1.

4.6.2 The terms employed in proof firing requirements contained herein are defined as follows:

The standard weight projectile weighs 33 lbs.

A standard round is one employing a standard weight projectile and loaded to give a muzzle velocity of 1,550 feet per second.

The rated maximum pressure is 32,000 pounds per square inch.

4.6.3 Powder used in rounds prescribed below by pressure shall be from a lot that gives the standard weight projectile a muzzle velocity of 1,550 feet per second with an average maximum powder pressure of not less than 29,000 nor more than 32,000 pounds per square inch. This restriction on powder is intended to eliminate excessive stresses in the mount and recoil mechanism during proof firing.

4.6.4 The first howitzer manufactured under any contract shall be proof fired in accordance with Table I.

TABLE I

Number of Rounds	Percent of Rated Maximum Pressure
1 or more	Standard Round
2	115
2	Standard Round
2	115
2	Standard Round

4.6.5 Each subsequent howitzer shall be proof fired in accordance with Table II.

TABLE II

Number of Rounds	Percent of Rated Maximum Pressure
1 or more	Standard Round
1 or more	115
1 or more	Standard Round

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4.6.6 The proof firing for acceptance of the howitzer may be combined with that of a mount and/or recoil mechanism. In such combined tests, rounds in addition to those required by the above tables may be fired.

4.6.7 The bore shall be star-gaged at the conclusion of the proof firing prescribed in Table I or Table II.

5. PREPARATION FOR DELIVERY

5.1 Preservation.-

5.1.1 Unless otherwise specified, all metal parts, before assembly, will be cleaned in accordance with Method C-2, as prescribed in Specification JAN-P-116.

5.1.2 Unless otherwise specified, before assembly, interior parts of the breech mechanism shall be preserved with a P-7 type preservative meeting the requirements of Specification JAN-P-116.

5.1.3 Unless otherwise specified, after assembly, the interior of the breech mechanism, interior of Howitzer tube and all exterior unpainted surfaces shall be preserved with a P-5 type preservative meeting the requirements of Specification JAN-P-116.

5.2 Packaging.- Unless otherwise specified, the Howitzer shall be packaged in accordance with Method I; the complete breech end shall be wrapped with a single thickness of Grade C, Type I greaseproof material meeting the requirements of Specification JAN-B-121.

5.3 Domestic shipment.- Unless otherwise specified, the Howitzer preserved and packaged as specified in 5.1 and 5.2 shall be packed in a suitable wood container so constructed as to insure safe delivery by common or other carrier to the point of delivery.

5.4 Overseas shipment.- Unless otherwise specified, the Howitzer preserved and packaged as specified in 5.1 and 5.2 shall be packed in a wood box meeting the requirements of JAN-P-106, Style 2 or 4. Each wood box shall be lined with case liner material meeting the requirements of Specification JAN-P-125. Closures shall be completely sealed with a continuous seam, at least 3/4-inch wide, of water-resistant adhesive complying with Specification JAN-P-140.

5.5 Marking.- Unless otherwise specified, shipping containers shall be marked in accordance with the requirements of Specification 100-2.

6. NOTES

6.1 Use.- This weapon is intended for installation in a self-propelled mount.

6.2 The circular advertisement and contract should definitely state the extra parts to be furnished with the howitzers. See paragraph titled "Extra parts" in section VII of Specification 51-1.

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